EMI Tcl commands

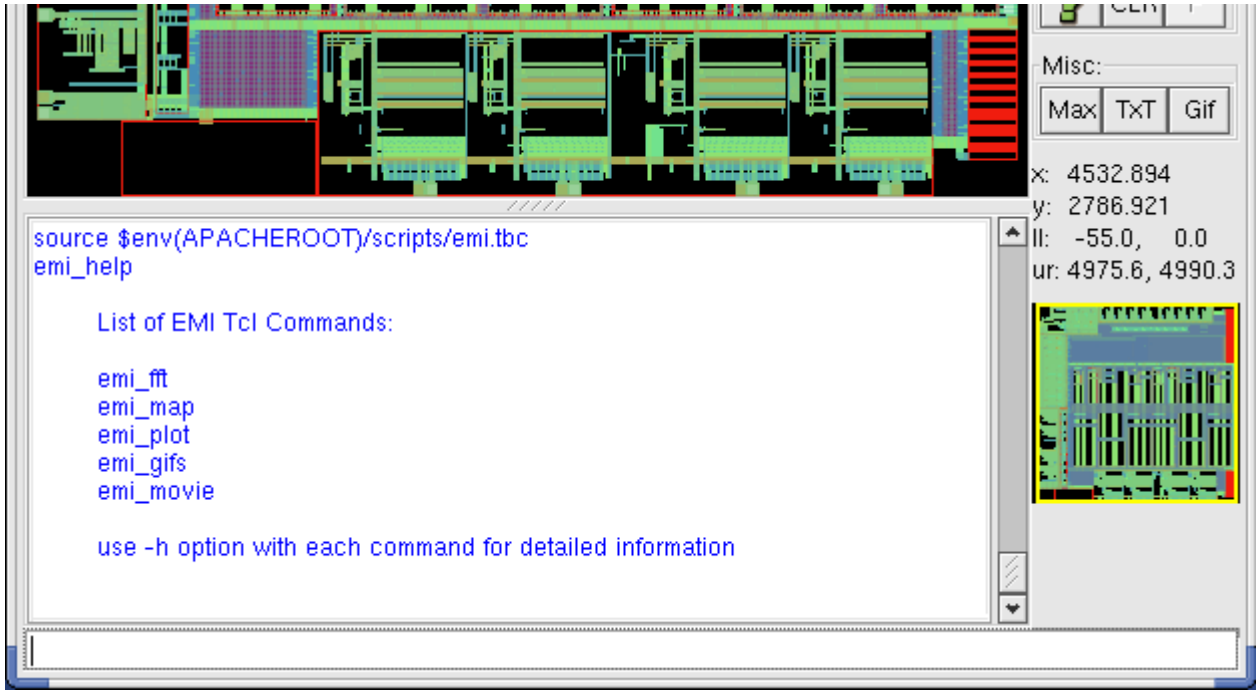
Apr., 2010

Apache Design Solutions

Loading emi.tbc

1. source $env(APACHEROOT)/scripts/emi.tbc [enter]
   - ✓ Valid in RH9.2.2 or newer version

2. emi_help [enter]
   - ✓ To see available commands



©2009 Apache Design Solutions

Partition current generation

- ‘–nx –ny –emi’ option is required in ‘perform analysis’
  - ✓ Partition number should be determined before analysis
    (ex) perform analysis –vcd –nx 20 –ny 20 –emi
  - ✓ Partition current **with** decap effect (partition ivdd) is generated as adsRpt/Dynamic/emi_ivdd.<domain>
  - ✓ Partition current **without** decap effect (partition ipwr) is generated as adsRpt/Dynamic/<design>.partition.<ivdd | ignd>
  - ✓ For partition ipwr, ‘–di’ option should be used in all commands
    (ex) emi_fft –di, emi_map –di, emi_plot –di

©2009 Apache Design Solutions

Running FFT

emi_fft [enter]

If error occurs, use a bit larger frequency step
(ex) emi_fft –step 400M [enter]

```
emi_fft
Info: Checking adsRpt/Dynamic/emi_ivdd.VDD
Info: Checking adsRpt/Dynamic/emi_ivdd.VSS
Info: Checking adsRpt/Dynamic/emi_ivdd.inst_129973_VDD_INT
Info: Removing previous temp data in fftTmp/
Info: Running FFT with freq step = 390.625MHz, max freq = 40.000GHz
Error: Please rerun a longer dynamic analysis or specify a larger minFreq. Exit
```

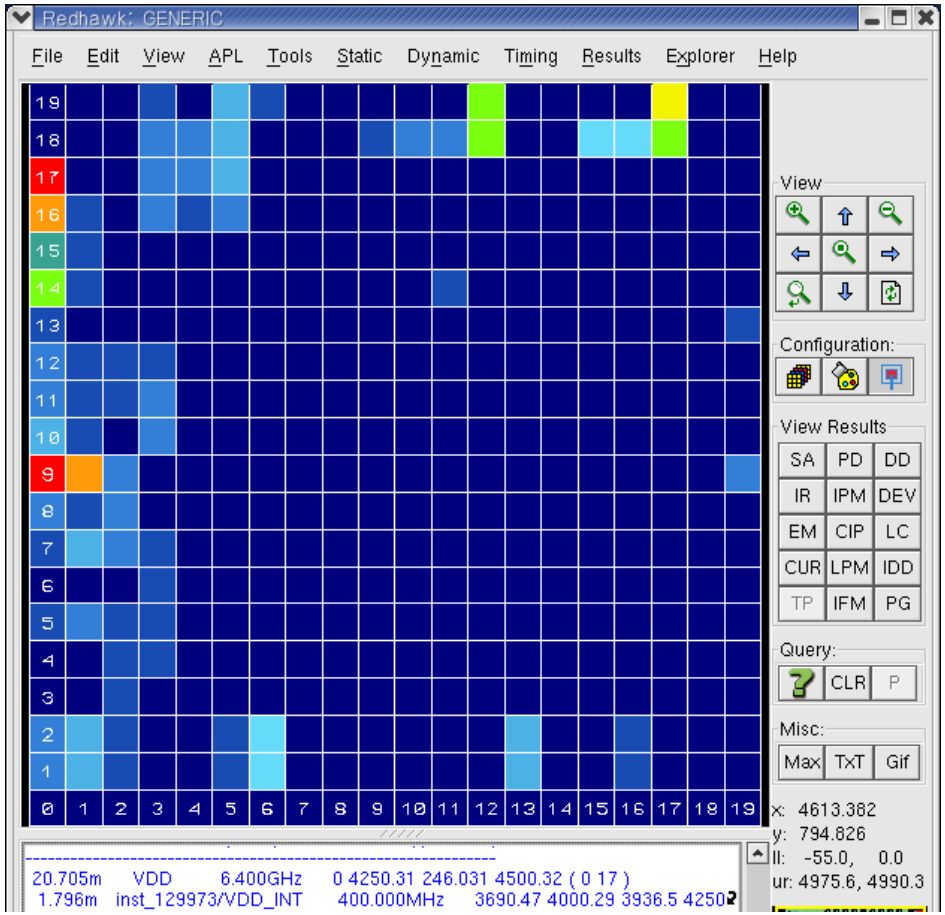
- Usage examples

emi_fft –min_f 1G –max_f 5G (to specify analysis frequency range)
emi_fft –net VDD –net VSS (to specify nets)
emi_fft –h (to see help)

©2009 Apache Design Solutions

Displaying EMI map (1/2)

emi_map [enter]



©2009 Apache Design Solutions

Displaying EMI map (2/2)

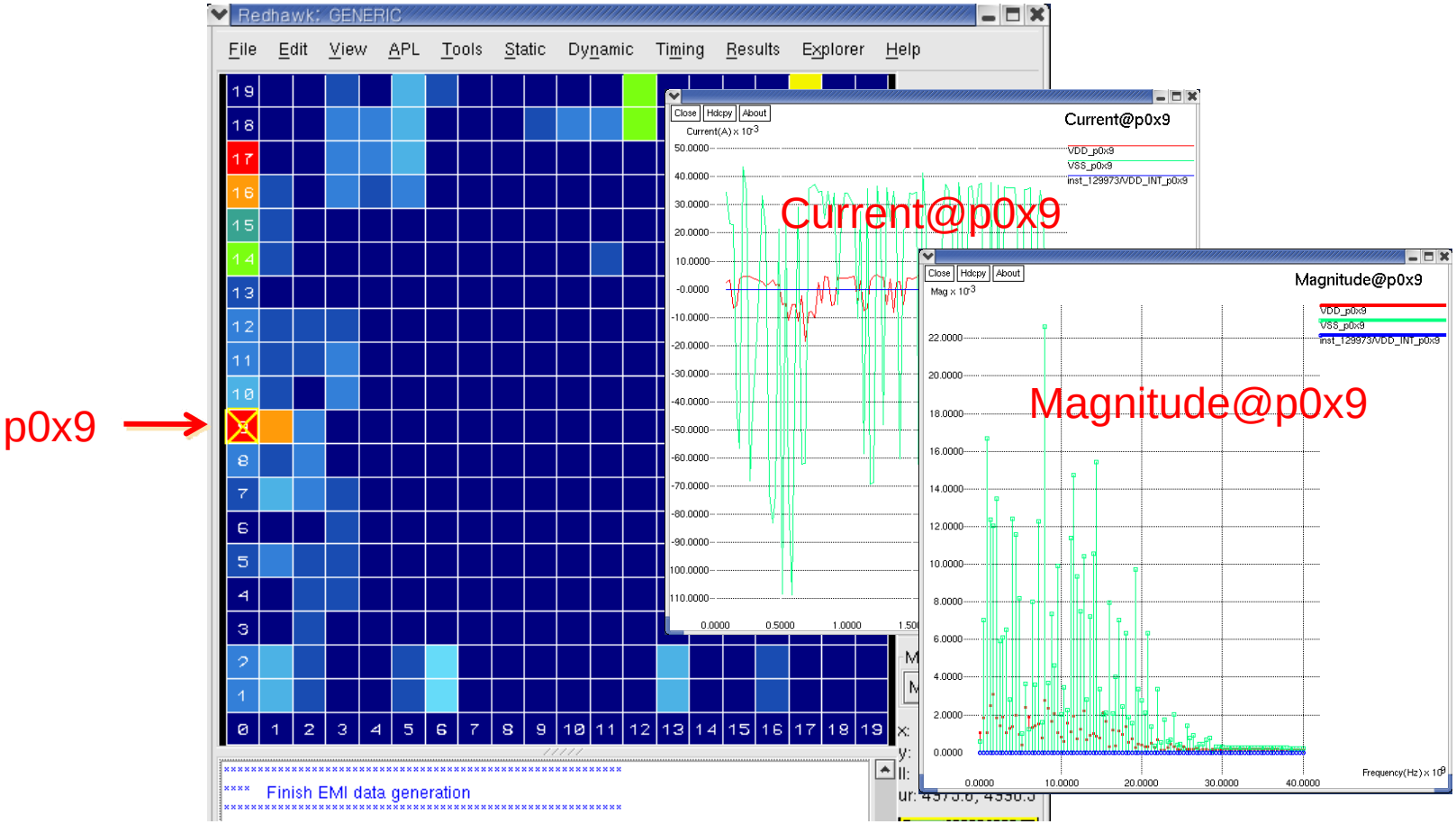
- Usage examples

  emi_map –f 2G (to see EMI map at 2GHz)
  emi_map –min_f 1G –max_f 5G (to set freq range)
  emi_map –min 10m –max 50m (to set color range)
  emi_map –net VDD –net VSS (to specify nets)
  emi_map –ng (to hide grid)
  emi_map –h (to see help)

©2009 Apache Design Solutions

Plotting spectrum/current (1/2)

emi_plot –p 0 9 [enter]



©2009 Apache Design Solutions

Plotting spectrum/current (2/2)

- Usage examples

  emi_plot –p 10 14 –p 7 8 (to plot multiple partitions)
  emi_plot –p 10 14 –s (to plot only spectrum)
  emi_plot –p 10 14 –c (to plot only current)
  emi_plot –p 10 14 –sv (to use SV)
  emi_plot –p 10 14 –net VDD –net VSS (to specify nets)
  emi_plot –p 10 14 –ng (to hide grid)
  emi_plot –h (to see help)

©2009 Apache Design Solutions

Other commands

emi_movie [enter]
emi_gifs [enter]

- Usage examples

  emi_movie –d 1 (to pause 1sec. between frames)
  emi_movie –min_f 1G –max_f 5G (to set freq range)
  emi_movie –step 100M (to set freq step)
  emi_movie –min 10m –max 50m (to set color range)
  emi_movie –h (to see help)

©2009 Apache Design Solutions

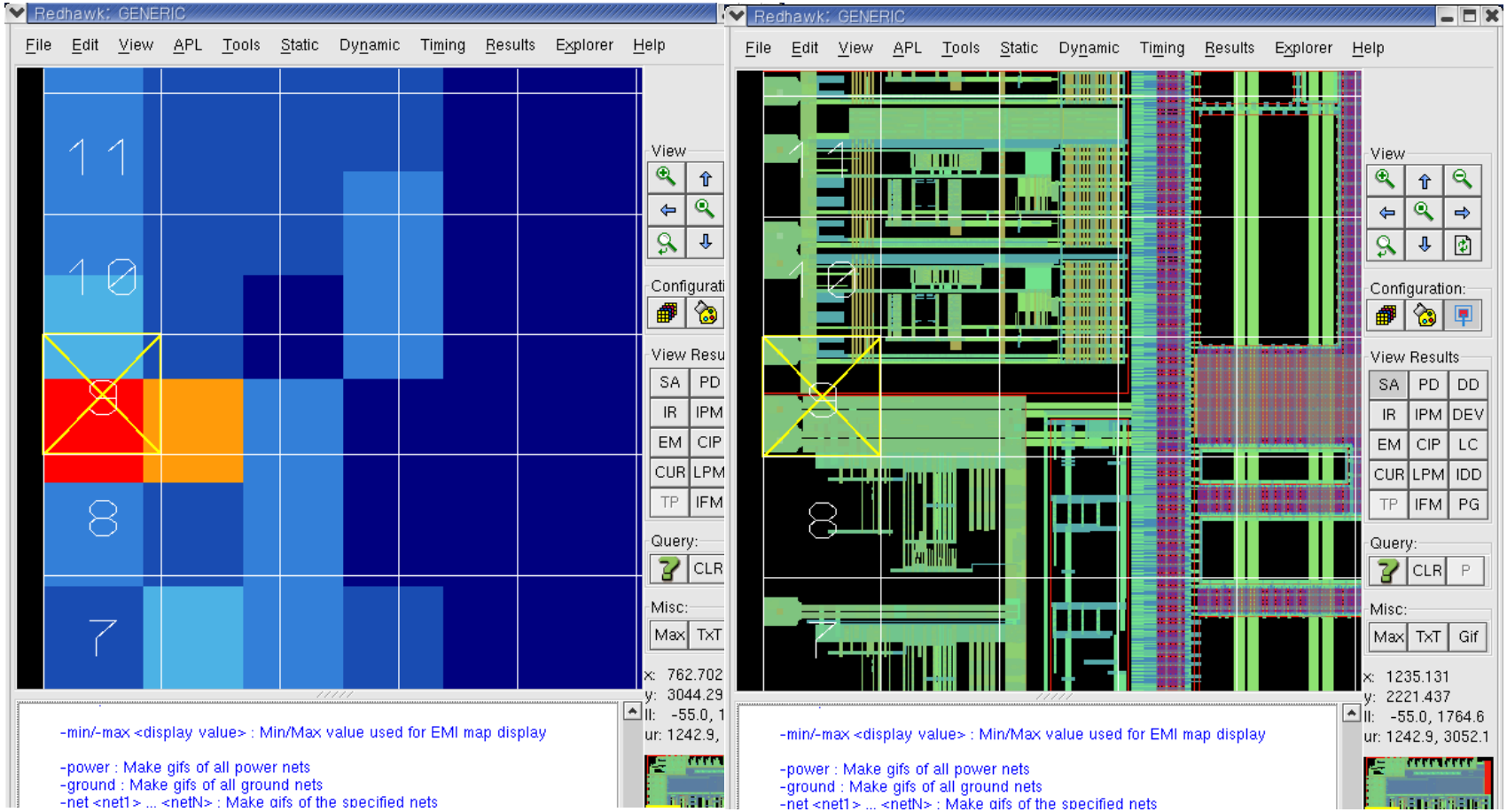
Directories and files

- fftTmp/ (fftTmp_di/)
  - ✓ data files from current2fft
- emiRpt/ (emiRpt_di/)
  - ✓ emi_fft.log (emi_fft log file)
  - ✓ report files (*.rpt)
- emiRpt/data/ (emiRpt_di/data/)
  - ✓ data files from emi_fft
- emiRpt/movie/ (emiRpt_di/movie/)
  - ✓ data files from emi_movie
- emiRpt/gifs/ (emiRpt_di/gifs/)
  - ✓ gif files dumped by emi_gifs

©2009 Apache Design Solutions

Known problem

When zoomed in, the tiles may be shifted by up to 1 tile size.
However, grids always indicate correct partition.



©2009 Apache Design Solutions